MD-88 Crew Audio Breakout Box

Rev C — Product Feature Specification

Overview: Battery-powered handheld audio mixing & communications box for MD-88 flight deck use. Connects between pilot headset and aircraft audio system. Provides mic protection, wireless crew comms (ESP-NOW), Bluetooth audio, ATC playback, real-time clock, and environmental sensors. Touch screen UI with Garmin-style control philosophy. Pilot-owned accessory, not FAA-certified — for training and personal cockpit use.

Core Audio Functions

Headset-to-Airframe Interface

- Connects pilot's GA headset (mic + headphones) to MD-88 airframe audio system
- Standard PJ-068 (¼" mic) and PJ-055B (0.206" phones) plug compatibility
- Captive pigtail to airframe-side jacks for vibration tolerance

Mic Protection

- Protects modern electret/dynamic mics from MD-88's intercom bias voltage
- R-C protection chain (4.7kΩ + 10µF AC-coupling)
- Auxiliary 10kΩ mic bias when off-airframe (for wireless intercom use)

Audio Output

- TPA6111A2 stereo headphone amplifier, 150mW per channel @ 16Ω
- 100µF polymer DC-blocking capacitors at output
- AC-coupled, no DC offset on headphone jacks
- MAX98357A I²S Class-D amp for audio processing path
- LC output filter for clean Class-D output

Wireless Communication

ESP-NOW Peer-to-Peer Intercom

- Wireless crew-to-crew audio between paired boxes (~100m line-of-sight)
- Low latency (~10ms typical)
- Paired ID provisioning at flash time + field re-pair sequence
- Designed for sets of 2 (pilot + copilot)

Bluetooth Music Input

- A2DP audio sink for streaming from GDL-52, iPhone, or other BT source

- Mixed with airframe audio or selectable mode
- Volume control via touch UI or encoder
- Mode: OFF / MIX / ONLY (selectable via menu)

WiFi Connectivity

- Connect to iPhone hotspot or aircraft WiFi
- OTA firmware updates
- Pulls METAR weather data from aviationweather.gov

User Interface

Touch Display

Hosyond 3.5" IPS color TFT, 320×480 resolution, capacitive touch input.

- Garmin-style menu navigation
- Battery icon with charging animation
- Volume bars and audio level meters
- Always-visible Zulu + Local time
- Cabin temperature display
- Configurable settings

Physical Controls

- EC11 rotary encoder with push button (volume, menu nav, turbulence backup)
- Tactile power button (Garmin-style: press to wake, hold 3s to power off)
- External PTT button connector (yoke-mounted button supported)
- External ATC playback button connector

Status Indicators

- Onboard charge LEDs (red = charging, green = full) — internal/service
- Panel status LED (firmware-controlled)
- Display-based status indicators (battery %, alerts, charging)

Power Architecture

Battery System

- 1500mAh LiPo pouch cell (803450 form factor, 8mm thick)
- ~10 hours runtime on battery (150mA average draw)
- TP4056 charger with DW01A + 8205A protection chain
- Soft-latch power switch (AO3401A P-FET) — true hardware off
- USB-C charging at 5V/1A
- Charges while box is on or off

Power Rails

Rail	Source	Voltage	Load
BAT+	LiPo via protection IC	3.0–4.2V	Always-on circuits
BAT_SW	AO3401A soft-latch output	3.0–4.2V	Switched main load
+5V	MT3608 boost from BAT_SW	5.0V	Audio amps, USB
+12V	MT3608 boost from BAT_SW	12.0V	LEMO Bose A20 power
+3V3	AMS1117 LDO from +5V	3.3V	Digital, sensors
USB_VBUS	USB-C VBUS pin	5.0V	TP4056 charger input

Power Button Behavior

- Press = power on (ESP32 boots, holds power via GPIO)
- Hold 3 seconds = soft power off (firmware saves state, releases hold)
- True hardware-off (no parasitic drain) for storage
- Reliable even with bricked firmware (button + P-FET work standalone)

Sensing & Monitoring

Environmental Sensor (BME280)

- Cabin temperature (±0.5°C accuracy)
- Humidity (0–100% RH)
- Barometric pressure (300–1100 hPa)
- Density altitude estimation
- Real-time display on touch UI

Real-Time Clock (DS3231SN)

- Zulu + Local time persistent across full power-off
- ±2ppm accuracy (drift <1 min/year)
- User-replaceable CR2032 coin cell (5–10 year life)
- Hardware alarm output for wake-on-time features

Other Inputs

- Battery voltage monitoring (ADC voltage divider)
- Mic level monitoring (ADC)
- Airframe audio monitoring (ADC)
- Audio jack detect for 3.5mm AUX input

Connectivity

USB-C Port (1×)

- Battery charging at 5V/1A
- Firmware programming via CH340C USB-UART bridge
- Auto-reset for esptool/Arduino IDE flashing
- Manual BOOT/RESET buttons for service mode

Audio Connections

- PJ-068 ¼" mono mic input (panel-mount jack)
- PJ-055B 0.206" stereo headphone output (panel-mount jack)
- 3.5mm AUX input (panel-mount jack)
- LEMO 6-pin Bose A20 (optional native ANC support)
- Mic + audio airframe pigtails (captive cables)
- External PTT button connector (JST PH 2-pin)
- External ATC playback button connector (JST PH 2-pin)

Software & Firmware

ProPilotApp iOS Integration

- BLE/WiFi pairing with iPhone
- Remote configuration via app
- Time sync from iPhone (when paired)
- METAR pull-through (app fetches, forwards to box)
- ATC playback library management
- Settings backup/restore

ATC Playback

- Pre-loaded clip library in WROVER 8MB flash (~30 min content)
- Single-button playback trigger
- OTA-updatable library via WiFi
- Mix or solo with live airframe audio

Firmware Updates

- USB-C via esptool / Arduino IDE (dev mode)
- WiFi OTA from ProPilotApp (user mode)
- Recovery via manual BOOT button (failsafe)

Audio Routing & DSP

- Master volume control
- Independent gain controls: mic, airframe, wireless intercom, BT music
- BT mode: OFF / MIX / ONLY
- Mute/unmute
- ATC playback compression (consistent loudness)

Physical & Mechanical

Enclosure

- ~98 × 55 × 30mm aluminum extrusion (Hammond 1455-class or similar)
- Black anodized aesthetic for cockpit ambiance
- Faceplate with display cutout + control mounts
- Panel + main PCB stacked vertically inside
- USB-C accessible from one edge

Mounting

- 4× M2 mounting holes per PCB
- Velcro or clip mount for cockpit attachment
- Compact form factor for kneeboard or yoke mounting

Reliability

- All JST PH SMD shrouded connectors (locking) for vibration
- All-SMD assembly (no through-hole vibration concerns)
- Industrial temperature parts where possible (-40 to +85°C)
- Conformal coating optional for production

Manufacturing

Item	Spec
Architecture	2-board: Main PCB + Panel daughter board
Inter-board cable	20-pin 0.5mm FFC, Type A, 100mm
Total components	~62 (50 main + 12 panel)
PCB layers	2-layer (4-layer optional for rev D)
Assembly	Full JLCPCB PCBA, all SMD
Prototype run	5 units (rev C)
Initial production	100 units (50 paired sets)
Estimated BOM cost	~$70–90 per unit at qty 100
Suggested retail	$200–300 per unit

Selling Points for Pilots

- **One box** between headset and airframe — clean and simple
- **Wireless crew comms** without yelling across cockpit
- **Bluetooth music** from any device
- **Save audio** for ATC review and study
- **Cabin environmental data** on-board (temperature, humidity, pressure)
- **Zulu time always available** — persists across power off
- **USB-C charge anywhere** — laptop, wall outlet, power bank
- **Aviation-grade durability** with locking JST connectors and vibration-resistant design
- **iPhone app integration** (ProPilotApp) for advanced configuration
- **Field-replaceable battery** (CR2032 for RTC backup)

Intentional Limitations

To set realistic expectations:

- **Not a radio** — relies on aircraft radio for transmit/receive
- **Not FAA-certified** — for pilot personal/training use only
- **Not flight controls** — audio and crew comms only
- **Not GPS/nav data** — display shows time and audio metadata only

Summary

The MD-88 Crew Audio Breakout Box is a battery-powered crew communication, audio mixing, reference clock, and sensor combo in a small handheld enclosure. It connects between the pilot's headset and the airplane's audio system, providing wireless intercom (ESP-NOW), Bluetooth music input, ATC playback, persistent Zulu time, and cabin environmental data — all without bulky panel-mounted hardware. Designed for the MD-88 cockpit but applicable to any GA aircraft with standard headset jacks.

Rev C beta production targets 100 units (50 paired sets) for distribution to professional pilots. Sourced from JLCPCB with full PCBA assembly.